PH3120 - Computational Physics Laboratory I

CPL109 - Chaos in Continuous Systems

1. Simple pendulum

Consider the equation for a simple pendulum, $\ddot{x} = \omega^2 x$.
This equation can be re-written as two first order differential equations.
$\dot{x}_1 = x_2$
$\dot{x}_2 = -\omega^2 x_1$

Assuming ω = 2, and initial conditions [0,1] for position and velocity respectively, solve the two first order differential equations.
Plot position and velocity as functions of time.
Plot the phase space diagram of velocity vs. time. Observe what happens to the phase space diagram when you give different initial conditions.
Plot the flow map of the simple harmonic motion for ω = 1, 2 and 3.

In Python, we can plot the flow of the differential equation for simple harmonic motion with the following code.

```
import numpy as np
import matplotlib.pyplot as plt
x = np.arange(-4, 4.1, 0.6)
y = np.arange(-6, 6.1, 0.81)
w = 2
xv, yv = np.meshgrid(x, y)
dx = yv
dy = -(w ** 2) * xv
mag = np.sqrt(dx ** 2 + dy ** 2)
dxu = dx / mag
dyu = dy / mag
plt.figure(1)
plt.quiver(xv,yv,dxu,dyu,scale=1,scale_units='xy',angles='xy',width=0.004,
color='b')
plt.xlabel('x')
plt.ylabel('y')
plt.title('Flow map for Simple Harmonic Motion')
plt.grid()
plt.show()
```

2. Damped pendulum

The equation for the damped pendulum is given by,

$$\ddot{x} = -\omega^2 x - \alpha\dot{x}$$

Write this equation as two first order differential equations. Solve the equations using python and plot the phase space diagrams for different values of α.
Include a few representative graphs in your report.

Modify the code to make a flow diagram for this case. Take $\omega = 1$ and $\alpha = 1$. Identify equilibrium points of this system. Expand the region near that point to see how the flow keeps pointing toward the equilibrium point. What type of a stable point is this?

3. In the damped harmonic oscillator, the resistive term alpha represents energy dissipation. If you make it negative, it will create energy.
   Make a flow diagram of this system. Identify the stable points and observe the behaviour of the flow near the stable point. What type of a stable point is this?

4. **Forced pendulum with damping**

   The equation for the forced pendulum with damping is,

$$\ddot{x} = -\omega^2 x - \alpha\dot{x} + A\cos\omega t$$

   Write this equation as two first order differential equations. Solve the equations and plot the phase space diagrams for alpha = 0.5, and A = 5. Try several initial conditions. Do you see an attractor here? If so, what type of an attractor it is?

5. The motion of a pendulum can be described by two first order differential equations. If x(*t*) represents the angle of a pendulum at time *t* and y(*t*) represents its angular velocity, the two equations will be,

$$y(t) = \frac{dx(t)}{dt}$$

$$\frac{dy(t)}{dt} = -\frac{g}{L}sin(x(t))$$

   Taking *g* = 9.81 and length *L* = 1, plot a flow map for the pendulum.

   This flow-map will give you a global view of the solutions of the differential equation with different initial conditions.

   There are two saddle node points in this diagram. To see one, expand the flow diagram to the region from 2 to 4 in x and from -2 to 2 in y. Observe how the flow happens near this saddle node, which is an unstable equilibrium point. Submit your diagram.

   Next, solve the differential equation assuming that the pendulum is initially held at an angle of $\frac{\pi}{6}$ and then released from rest. Plot a graph of y vs. x.

   After that, repeat this for initial angles with values such as $\frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}, \frac{3\pi}{4}, \frac{5\pi}{6}$ and $\frac{7\pi}{6}$.

   Then plot the same graph for any initial angle, but with a large velocity (such as y = 10).

   Submit your flow map and other plots. Also include answers to the following questions.

   a) Under what conditions do you get periodic trajectories?
   b) What kind of trajectories do you get when the pendulum is released at rest, from any initial angle less than π ?

c) What kind of trajectories do you get when the pendulum is released at rest, from any initial angle greater than π ?
d) What kind of trajectories do you get when the initial angular velocity y is very high?

6. Van der Pol oscillator is a dynamical system with a non-linear damping. It is described by the following differential equation.

$$\frac{d^2x}{dt^2} - \mu(1 - x^2)\frac{dx}{dt} + x = 0$$

Taking $\mu = 1$ and using initial conditions (1,1), solve this equation and plot its trajectory. You will find that the trajectory is a closed curve.

Now make a flow diagram of the Van der Pol oscillator and plot the above trajectory on the same diagram. You will see that, from outside the loop or from inside the loop, the flow is always pointing toward the trajectory. That means, this closed trajectory is a stable one. This is called a limit cycle. Submit your diagram.

<u>Lorentz System</u>

7. Lorentz system is described by the following set of differential equations.

$$\frac{dx}{dt} = \sigma(y - x)$$
$$\frac{dy}{dt} = x(\rho - z) - y$$
$$\frac{dz}{dt} = xy - \beta z$$

Solve the Lorenz system and plot the phase space diagrams for the following values of constants.

ρ	σ	β
14	10	8/3
15	10	8/3
28	10	8/3

The last set of parameters shown above will produce chaos. Because this is a three-dimensional system, you can plot two-dimensional projections.
Plot x vs. y and y vs. z graphs. You can also try making three-dimensional plots.
What happens when you change initial conditions?
Make plots of the variables as a function of time. What sort of graphs do you get when the system is chaotic?
Submit your code with several graphs.

Sensitivity to initial conditions

7. Plot a graph of *x* vs. time for the following values of constants, with initial conditions x=0, y=1 and z=1. Make sure that you do the calculation from *t*=0 to about *t*=20 or 30.

ρ	σ	β
70	10	8/3

On the same graph, plot the trajectory with the same parameter values, but with initial conditions x = 0.001, y=1 and z=1.

Observe the difference between the two trajectories that result from this slightly different initial conditions. Try using initial conditions differing only by about 0.0000001. Do you still see the trajectories diverging with time?

What you observe here is called extreme sensitivity to initial conditions (or chaos).

Submit the code used with one graph showing sensitivity to initial conditions.

Stable Points

8. For the following parameter values the Lorenz system is not chaotic.

ρ	σ	β
10	10	8/3

It is possible to show that the system has three stable points under this condition. One of them is at (0,0,0). The other two are near x=5, y=5 and x = -5, y = -5.

By plotting phase space diagrams with suitable initial conditions, try to find these stable points. Also determine the value of z for the two points not on the origin.

Submit your graphs showing the stable points.

<u>Duffing Equation</u>

9. Solve the following equation for parameter values described in the link given below.

$$\ddot{x} + \delta\dot{x} + \alpha x + \beta x^3 = \gamma\cos(\omega t)$$

https://en.wikipedia.org/wiki/Duffing_equation#Examples

10. Hyper-chaotic behavior, strange behavior,

- Lyapunov exponents (λ) were identified at CPL 108. More than one positive Lyapunov exponent indicates the presence of hyperchaos. There, chaos has divergence in multiple directions

$$\lambda = \lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} \ln|f'(x_n)|,$$

- Given ordered Lyapunov exponents $\lambda_1 \geq \lambda_2 \geq \lambda_3$, the Kaplan-Yorke Fractional dimension is,
  $D_{KY} = j + \sum_{i=1}^{j} \lambda_i \ /|\lambda_{j+1}|$, j is the largest integer $\sum_{i=1}^{j} \lambda_i \geq 0$.

  This dimension estimates how many effective degrees of freedom a system uses on its attractor. In a three-dimensional system, if this dimension is between 2 and 3, it indicates a strange behavior. It is because these trajectories are not confined to a simple 2D surface. But also don't fill a 3D volume entirely. Instead, they lie on a fractally folded surface, which is a characteristic of a strange attractor.

- Use the Lorentz system with the following parameters, $\sigma = 10, \rho = 28, \beta = \frac{8}{3}$, and initial conditions $x_0 = 1.0, y_0 = 1.0, z_0 = 1.0$. Then calculate the Lyapunov exponents of the system. Then calculate the Kaplan-Yorke Fractional dimension, and finally reveal whether the system has hyperchaotic or strange behavior.